

EMPLOYMENT

Massachusetts Institute of Technology
 C. L. E. Moore instructor, Mentor: Wei Zhang

Cambridge, MA
 Fall 2026 - Spring 2029

EDUCATION

University of Wisconsin - Madison

Ph.D. in Mathematics, Advisors: Michael Rapoport (primary), Tonghai Yang (secondary)

Madison, WI

May 2026

University of Wisconsin - Madison

M.A. in Mathematics

Madison, WI

May 2021

Zhejiang University

B.S. in Mathematics, Advisor: Fei Yu

Hangzhou, ZJ

May 2020

PREPRINTS

1. Yu Luo, “Kudla–Rapoport conjecture for unramified maximal parahoric level”, [submitted](#) .54pp
2. Yu Luo, Andreas Mihatsch and Zhiyu Zhang, “Unitary Shimura varieties at ramified primes and arithmetic transfer”, [submitted](#) .73pp
3. Yu Luo, Michael Rapoport and Wei Zhang, “More regular formal moduli spaces and arithmetic transfer conjectures: the ramified quadratic case”, [submitted](#) .98pp
4. Qiao He, Yu Luo and Yousheng Shi, “The basic locus of ramified unitary Rapoport–Zink spaces at maximal vertex level”, [submitted](#) .69pp
5. Qiao He, Yu Luo and Yousheng Shi, “Regular models of ramified unitary Shimura varieties at maximal parahoric level”, [submitted](#) .41pp

PUBLICATIONS

1. Yu Luo, “On the moduli description of ramified unitary local models of signature $(n - 1, 1)$ ”, **Math. Ann.** volume 392, pages 4661–4738 (2025)

INVITED TALKS

- Arithmetic Siegel–Weil formula and Kudla–Rapoport conjecture June 2026
Fudan Workshop on Number Theory and Arithmetic Geometry, Shanghai, China
- Arithmetic Siegel–Weil formula at maximal parahoric level April 2026
Caltech Number Theory Seminar, Pasadena, California, USA
- Geometric and arithmetic Siegel–Weil formula Feb 2026
Johns Hopkins Junior Number Theory Days, Baltimore, Maryland, USA
- A new proof of the arithmetic Siegel–Weil formula Jan 2026
Joint Mathematics Meetings 2026, Washington, D.C., USA

- On unitary Rapoport–Zink space Nov 2025
Universität Münster, Münster, Germany
- Proof of the arithmetic Siegel-Weil formula Nov 2025
University of Duisburg-Essen, Essen, Germany
- A new proof of the arithmetic Siegel-Weil formula Nov 2025
Universität Münster, Münster, Germany
- A new proof of the arithmetic Siegel-Weil formula Sept 2025
POSTECH, Pohang, South Korea
- A new proof of the arithmetic Siegel-Weil formula June 2025
Symposium on Number Theory and Representation Theory, IX, Hangzhou, China
- A new proof of the arithmetic Siegel-Weil formula May 2025
Zhejiang University
- A new proof of the arithmetic Siegel-Weil formula Feb 2025
University of Wisconsin - Madison
- Comparison of absolute and relative unitary Rapoport-Zink spaces Oct 2024
Massachusetts Institute of Technology (Learning seminar on Arithmetic Inner Product Formula)
- On the Moduli Description of a Class of Unitary Shimura Varieties Dec 2023
University of Wisconsin - Madison
- On the Moduli Description of a Class of Unitary Shimura Varieties May 2023
Zhejiang University
- On the Moduli Description of a Class of Unitary Shimura Varieties May 2023
Michigan State University

SERVICES

- Co-organizer of the Langlands–Kottwitz method reading Seminar 2025 Fall - 2026 Spring
University of Wisconsin - Madison
- Co-organizer of the Relative trace formula reading Seminar 2024 Fall - 2025 Spring
University of Wisconsin - Madison
- Organizer of the Gross–Zagier formula reading Seminar 2024 Spring
University of Wisconsin - Madison
- Co-organizer of GAGS - Graduate Algebraic Geometry Seminar 2022 Spring–2025 Spring
University of Wisconsin - Madison
- Mentor at Undergraduate Directed Reading Program 2022 Spring
Mentor of Baoyi WANG (Currently in Columbia University) University of Wisconsin - Madison
- Mentor of Master Program in Math Department 2021 Fall
Mentor of Ruoxi LI (Currently in UIUC Math PhD), Weiman YUAN (Currently in Vanderbilt University Bio, Engineering), Aditya Phukon.
- Organizer of Algebraic Geometry Learning Seminar 2020 Spring
University of Wisconsin - Madison
- Organizer of Commutative Algebra Learning Seminar 2019 Fall
University of Wisconsin - Madison
- Organizer of Undergraduate Algebraic Geometry Seminar 2019 Spring
Zhejiang University

TEACHING

- Math 340 - Linear algebra - TA 2026 Spring
University of Wisconsin - Madison
- Math 221 - Calculus 1 - TA 2025 Fall
University of Wisconsin - Madison
- Math 475 - Introduction to Combinatorics - TA 2025 Summer
University of Wisconsin - Madison
- Math 435 - Introduction to Cryptography - TA 2025 Summer
University of Wisconsin - Madison
- Math 340 - Linear Algebra - TA 2025 Spring
University of Wisconsin - Madison
- Math 112 - Algebra - Lecturer 2024 Fall
University of Wisconsin - Madison
- Math 320 - Linear Algebra and ODE - TA 2024 Spring
University of Wisconsin - Madison
- Math 340 - Linear Algebra - TA 2023 Fall
University of Wisconsin - Madison
- Math 340 - Linear Algebra - TA 2023 Spring
University of Wisconsin - Madison
- Math 234 - Calculus 3 - TA 2022 Fall
University of Wisconsin - Madison
- Math 240 - Discrete Math - TA 2024 Summer
University of Wisconsin - Madison
- Math 222 - Calculus 2 - TA 2022 Spring
University of Wisconsin - Madison
- Math 221 - Calculus 1 - TA 2021 Fall
University of Wisconsin - Madison

HONORS AND AWARDS

- ICM 2026 Travel Grant Program 2026 Summer
- Excellence in Mathematical Research Award - University of Wisconsin, Madison 2025 Fall
- Government Scholarship - Zhejiang Government 2017–2018
- Government Scholarship - Zhejiang Government 2016–2017